IoT Integrated Machine Learning Model for Poultry Gender Prediction Based on egg morphometric.

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Introduction

The poultry industry faces a major ethical and operational challenge in identifying chick gender only after hatching, leading to the widespread culling of male chicks[1]. This project proposes an innovative solution using IoT, computer vision, and machine learning to predict gender before incubation. By analyzing egg dimensions and calculating the shape index in real time, the system enables early, non-invasive gender detection helping reduce waste, improve efficiency, and support sustainable, ethical poultry farming practices[2],[6].

Integrating IoT and machine learning offers a non-invasive, cost-effective solution for early gender prediction, optimizing resources and mitigating ethical concerns[1].

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Problem Statement

The poultry industry struggles with ethical and efficient gender identification. Male chicks, unsuitable for egg or meat production, are culled after a 21-day incubation, wasting resources[1],[4].

Current methods are either time-consuming and error-prone (manual) or expensive and slow (laboratory-based), making them impractical for large-scale operations. A rapid, non-invasive, and cost-effective solution is urgently needed[3],[6].

Research Questions

1. How can computer vision techniques be used to accurately measure egg morphometric traits such as width and height for early gender prediction in poultry farming?

2. What is the most accurate and computationally efficient machine learning model for classifying poultry gender based on non-invasive egg morphometric data in real-time?

3. To what extent does the integration of IoT and machine learning reduce operational costs, resource wastage, and ethical concerns in commercial poultry hatcheries compared to traditional gender identification methods?

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Research Objectives

Develop IoT Model

Create an IoT system to collect egg morphometric data (e.g., Egg Shape Index) using hatchery devices.

Use Computer Vision AI

Implement object detection techniques for accurate egg dimension measurement.

Develop ML Model

Build a robust machine learning model to improve gender prediction accuracy.

Develop Mobile App

Create a user-friendly mobile application for clear viewing of results and informed decision-making.

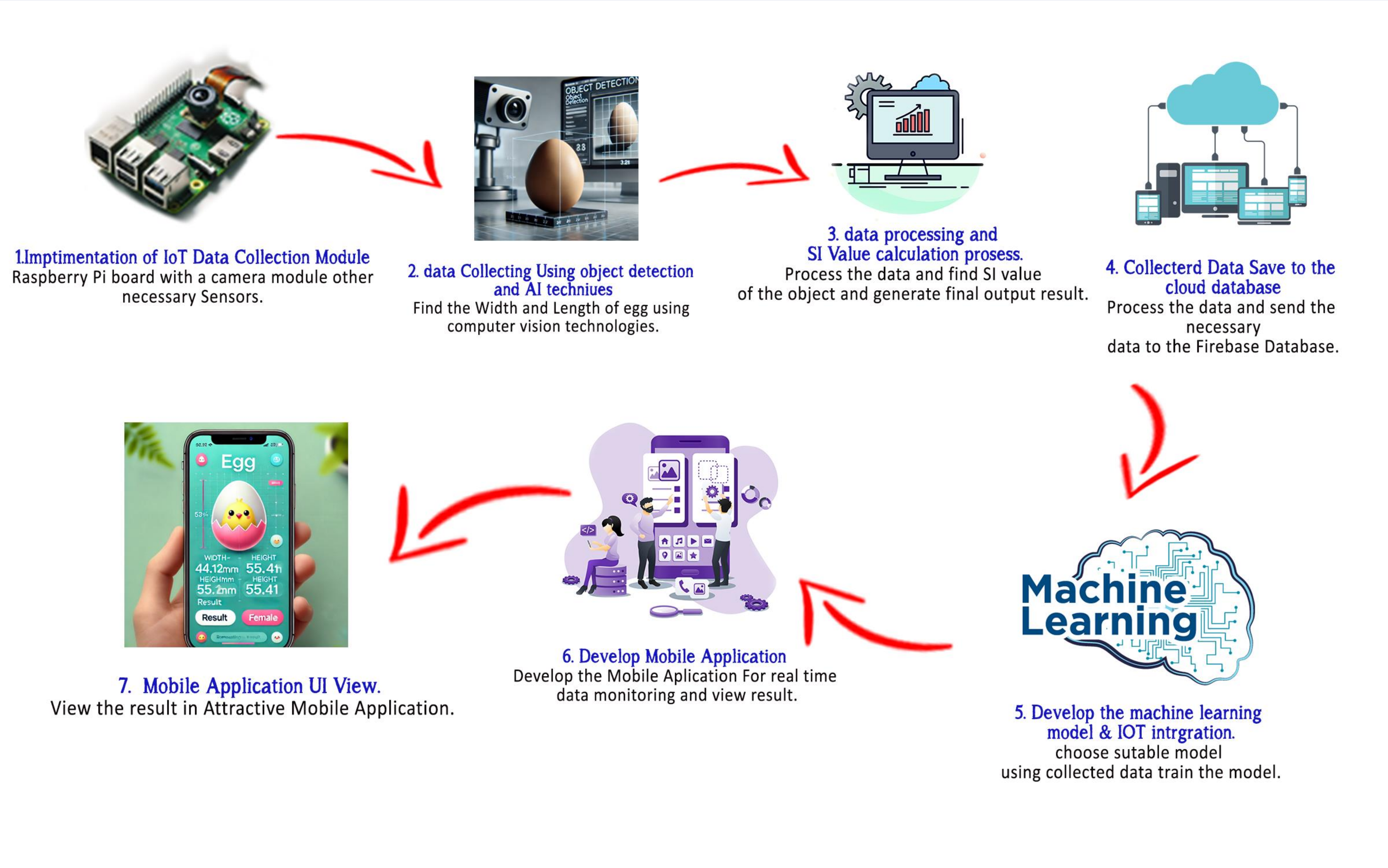
Literature Review.

Traditional chick sexing methods, including vent sexing and feather observation, are widely used but are manual, invasive, and only applicable post-hatching, making them inefficient and unsuitable for large-scale hatcheries [5]. Laboratory-based approaches such as DNA testing and spectroscopy provide greater accuracy but are costly, time-consuming, and not feasible for real-time industrial applications [1],[4].

As a non-invasive alternative,recent studies have focused on egg morphometric features—particularly width, height, and the Egg Shape Index (ESI)—as indicators for early gender prediction[5]. Research shows a statistically significant correlation between ESI values and chick gender, with certain shapes more likely to result in female embryos [3].

IoT-enabled devices such as Raspberry Pi, camera modules, and sensors support real-time monitoring and cloud-based data transmission, reducing labor and increasing efficiency [5], [6]. Moreover, machine learning models such as Support Vector Machines and Random Forest classifiers have demonstrated high accuracy in gender classification when trained on morphometricdata, proving the viability of a scalable, automated, and ethical system for early poultry gender prediction [2], [5].

Methodology



1.Plan and IoT System, Sensor Selection, and Workflow

The project's IoT system is designed to automate the acquisition of egg morphometric traits specifically width and height for early gender prediction using machine learning.

Sensor and Hardware Selection:

Component	Purpose
Raspberry Pi 4 Model B	Acts as the central controller and ML inference engine.
USB Camera Module	Captures high-resolution images of eggs for morphometric analysis.
LED Ring Light	Provides uniform illumination to eliminate shadows during image capture

2.Machine Learning Model Development

The machine learning model serves as the core intelligence layer of the system, responsible for predicting the gender of chicken embryos using measurable egg morphometric features specifically width, height, and the Egg Shape Index (ESI).

Several ML models can be used to classify chicken egg gender based on width, height, and Egg ShapeIndex (ESI):

- **Random Forest:** Highly accurate, robust, and handles complex data well. Ideal for your main model.
- **Support Vector Machine (SVM):** Great for binary classification with small datasets; needs normalized data.
- **Logistic Regression: Simple,** interpretable, and good as a baseline model.
- **Decision Tree:** Fast and easy to understand; useful for quick testing.
- **K-Nearest Neighbors (KNN):** Simple but less efficient for large datasets.

3.Mobile App Design and Development

1. Real-Time Monitoring: The app displays live data such as egg width, height, ESI, and predicted gender directly from the IoT system.

2. User-Friendly Interface: Designed with a simple UI for hatchery staff to easily view results, trends, and make quick incubation decisions.

3. Cloud Integration: Connected to Firebase or a cloud database for seamless data sync, storage, and remote access

- Android studio,
- Programming languages (dart/python)
- Database (Firebase)
- Cloud services (Firebase)
- Machine learning environment (Microsoft Azure/Latent AI)
- Software Development Tools (VS Code)
- Documentation (Ms Word)

3. Budget & Resourses.

Component	Budget
Raspberry Pi 4 Model B	Rs.25000.00
USB Camera Module	Rs.2500.00
LED Ring Light	Rs.900.00
Cables and Enclosure Box	Rs.5500.00
Total Cost	Rs.33900.00 /=

3. TimeLine

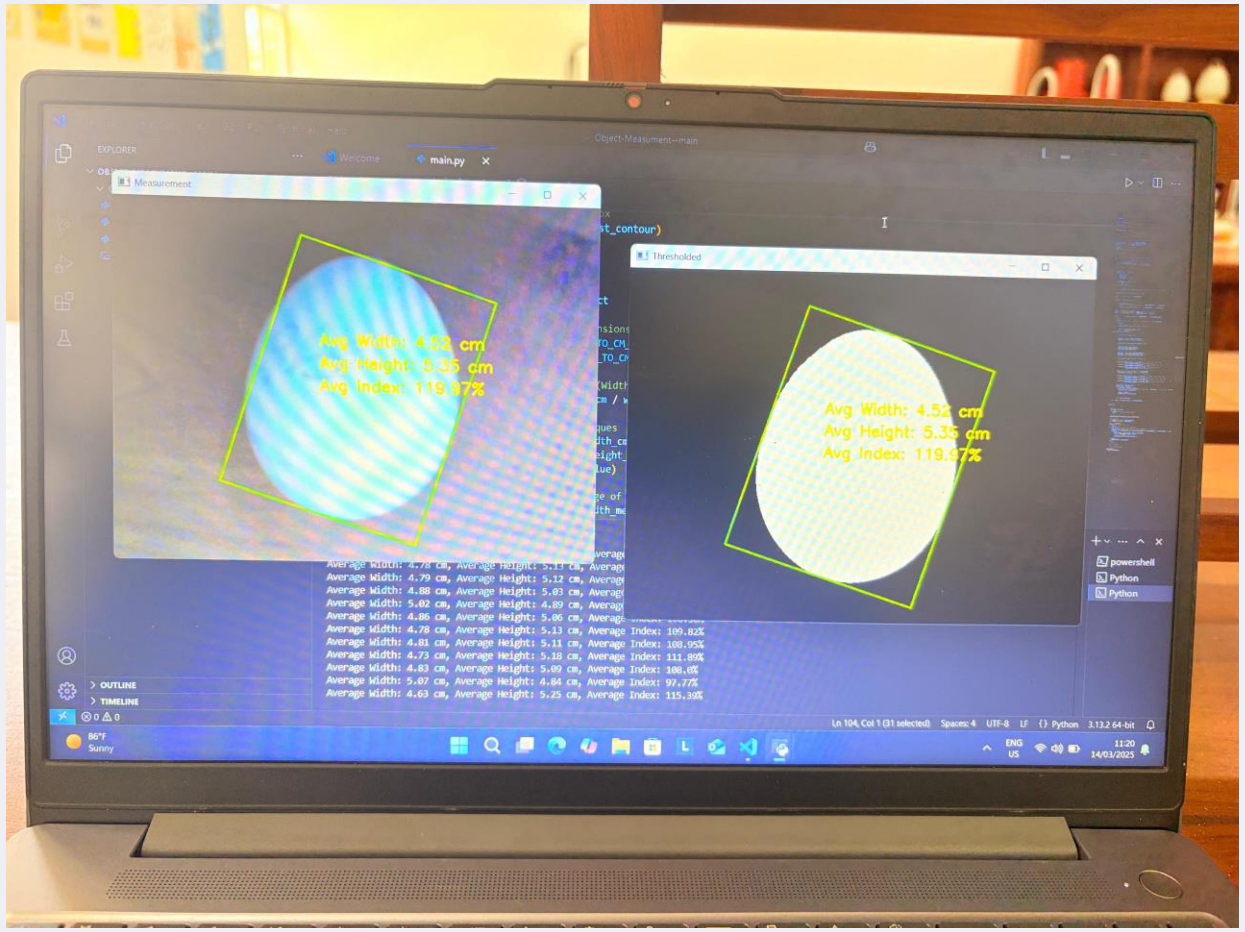
	2024						2025							
OBJECTIVES	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	March	April	May	June	July	Aug
Basic Literature Review														
Proposal Submission														
Literature Review														
IoT Device Simulation & Planning														
Component Gathering and Building Device														
Model Creation & Tuning														
Develop Mobile Application														
Application Testing														
Thesis Proof Reading														
Writing the Thesis														

Implementation

Work Completed:

- Data Collection Initiated: Collected initial egg samples for morphometric analysis.
- Prototype Built: Developed a basic prototype using usb camera and laptop to collect egg morphometric data.
- Morphometric Reading via OpenCV: Implemented OpenCV to extract egg width and height
- Egg Shape Index Calculation: Calculated ESI using width and height measurements.
- Dataset Preparation: Structured and labeled dataset created with width, height, ESI, and corresponding gender labels for ML training.

Work Completed:



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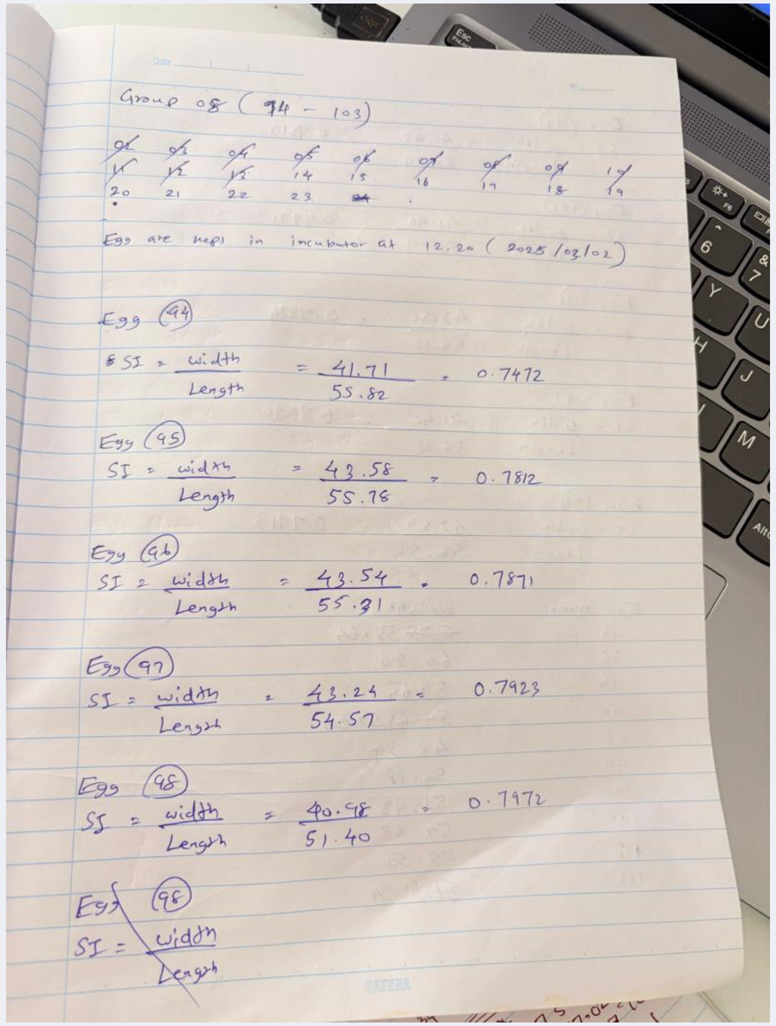
Work Completed:



Work Remaining:

- Develop the machine learning model with collected data set to get accurate results.
- Refinement and optimization of ML models for higher accuracy.
- Develop the mobile application.
- Develop final project using Raspberry Pi board and other equipment.

References:

[1] M. Kayadan and Y. Uzun, “High accuracy gender determination using the egg shape index,” Scientific Reports, vol. 13, no. 504, 2023.

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[3] V. U. Oleforuh-Okoleh, “Hatchability Prediction in Chickens Using Some External Egg Quality Traits,” Asian Journal of Animal Sciences, vol. 10, no. 2, pp.159–164, 2016.

[4] M. Narushin and M. Romanov, “Egg physical characteristics and hatchability,” World’s Poultry Science Journal, vol. 58, no. 3, pp. 297–303, 2002.

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[6] U. Asil and E. Nasibov, “Sex Detection in the Early Stage of Fertilized Chicken Eggs via Image Recognition,” International Journal of Computer Science and Information Technology, vol. 15, no. 2, pp. 19–28, 2023.

THANK YOU !